Neel Shrivastava

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EDUCATION

University of Wisconsin-Madison Expected May 2029
B.S. in Biomedical Engineering and Minor in Computer Science *Madison, WI*

- Specialization: Biomedical Engineering on Bioinstrumentation and Electronics track
- Coursework: Engineering Design 170, Calculus 1 & 2, Dif Eq & Linear Algebra, Physics: Mechanics, AP CS: Java.
- Awards: IBM Thomas J Watson Scholarship recipient, Wayzata Debate Scholarship recipient (2% reception)

EXPERIENCE AND PROJECTS

Wisconsin MakerSpace August 2026 – Present
Biomedical Engineer *Madison, WI*

- Designing & integrating hardware solution using load sensors, ADC's, wiring, NFC readers/stickers, and ESP32 microcontroller with Python & C firmware to track precise material usage in 3d printers.
- Calibrating analog sensor inputs into digital telemetry data, transmitting real-time stock levels over Wi-Fi to automated inventory database.
- Designing climate controlled storage in CAD and OnShape, and fabricating with saws, lasers, and 3D printing.

Silicon Badgers | *Verilog, Chip design, logic* August 2026 - Present

- Wrote SystemVerilog RTL and supporting code to build ISA calculator, and tested using QuestaSim
- Gained experience in full ASIC and FPGA flow to learn software-chip interactions and chip logic/design
- Learned how neural networks use INT8/INT32 data types, memory mapping, and Multiply-Accumulate operations
- Learned computer architecture concepts such as instruction decoding, datapath operations, and state machines

Wisconsin FSAE Racing (Firmware Team) | *C, Microcontrollers, GPIOs* August 2026 - Present

- Completed embedded C onboarding on ESP32/Arduino MCUs; foundation for team steering & control software
- Configured/controlled GPIO pins to read inputs & drive hardware output; learning firmware-physical interactions
- Used Git for version control in team codebase, with branching & committing through team GitLab workflow
- Currently developing temperature sensor firmware on ESP32 to read & process real data

Medical Text Reader | *API, Python* July 2026

- Built medical text processor that translates complex medical text and prescriptions into plain, patient-friendly language. Created file upload functionality with R.
- Built Python-based backend, integrating Groq Llama 3 and Gemini APIs for image processing and analysis

Run Protect | *Python, HTML, CSS* April 2025

- Developed a web app that predicts running-related overuse injuries from biomechanical data using a rule based composite scoring algorithm in Glide.
- Built UI/UX using HTML, CSS, JavaScript in React and Tailwind, facilitating 100+ users on 10x retention rate
- Engineered a robust backend using Python and Java, integrating Firebase to securely collect, structure, and query real-time biomechanical data

Housing Price Simulation Model | *Python, Prediction Modeling* June 2026

- Developed a house price model that simulates real wage growth, inflation, and estate prices to recommend a purchase timeline. Utilized real data and averages.
- Conducted Stochastic Return Modeling through various simulations across multiple financial models with list and dictionary storage.

ADDITIONAL INFORMATION

Skills : Python, Java, CAD, OnShape, Solidworks, GitHub, GitLab, Codex, VS Code, Public speaking/Presentation
Clubs: Wisconsin Racing (Firmware), UW BME society, Silicon Badgers (Chip design), UW Transcend
Interests: Spiderman Films, Ed Sheeran, Premier League Soccer, Manchester City FC, LA Rams, UW Badgers